



The University of Manchester

Pretraining Large Language Models

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Learning Outcomes

- **Define LLM pretraining** and explain why it is **self-supervised**.
- **Compare Transformer variants** (encoder-only, encoder–decoder, decoder-only) and explain why modern LLMs use **decoder-only with causal attention**.
- **Formulate the training objective**: autoregressive probability factorisation, **cross-entropy loss**, and the difference between **training vs inference**.
- **Evaluate implications**: limitations (e.g., hallucinations/bias), the impact of **scale/compute**, and how pretraining connects to **adaptation and instruction tuning**.

Where We Are in the Pipeline

- Previously:
 - Transformers
 - BERT
 - Training data for LLMs
- Today:
 - How large language models are pretrained

Transformer

Encoder Only

Uses: NLU, classification, feature extraction

Examples: BERT, RoBERTa, DeBERTa

Encoder-Decoder

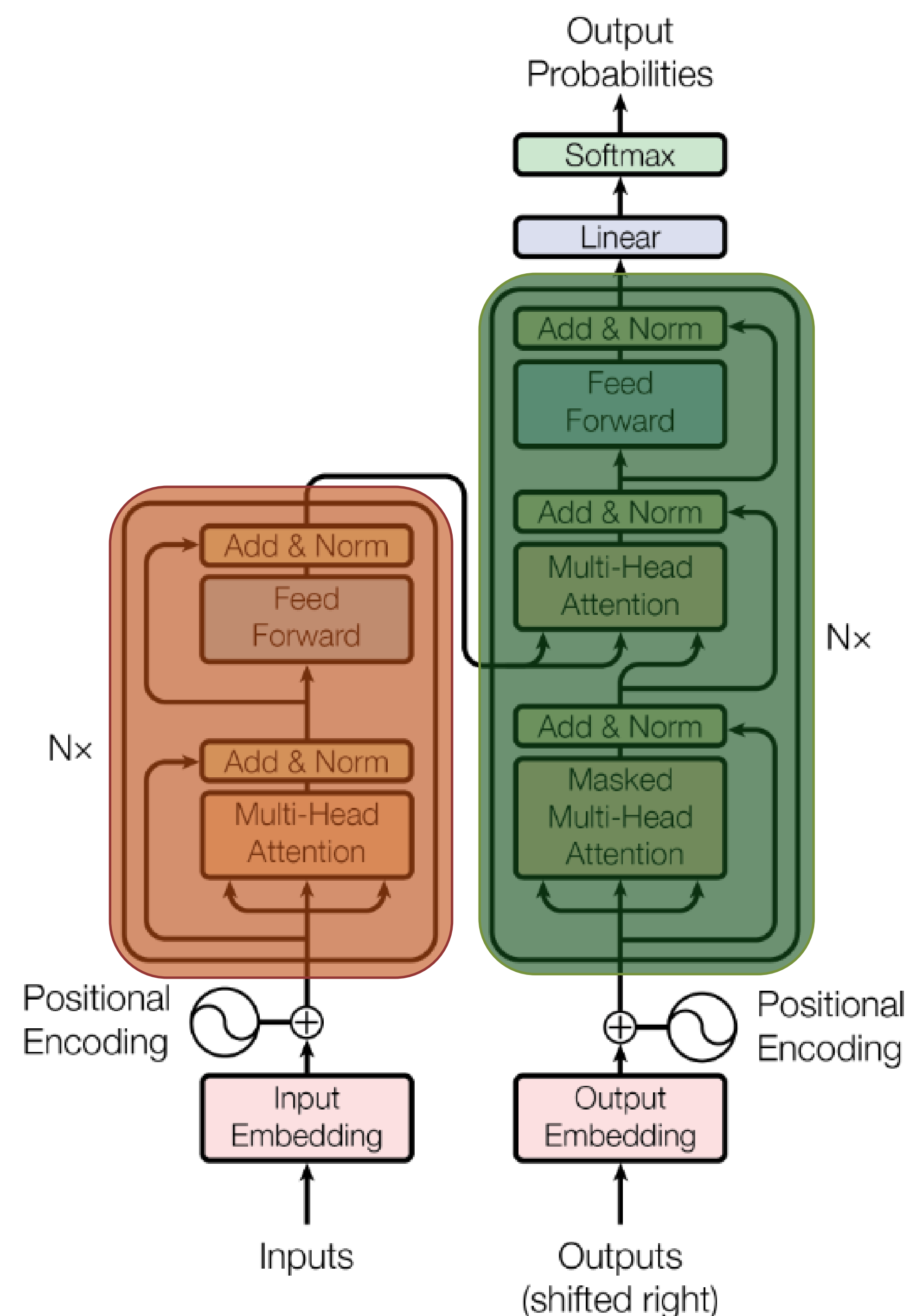
Uses: NLG, translation, summarisation, seq2seq mapping

Examples: T5, Flan-T5, BART

Decoder Only

Uses: NLG, translation, summarisation, completion

Examples: GPT-x, LLaMa, every other new model today



Decoder-Only Architecture

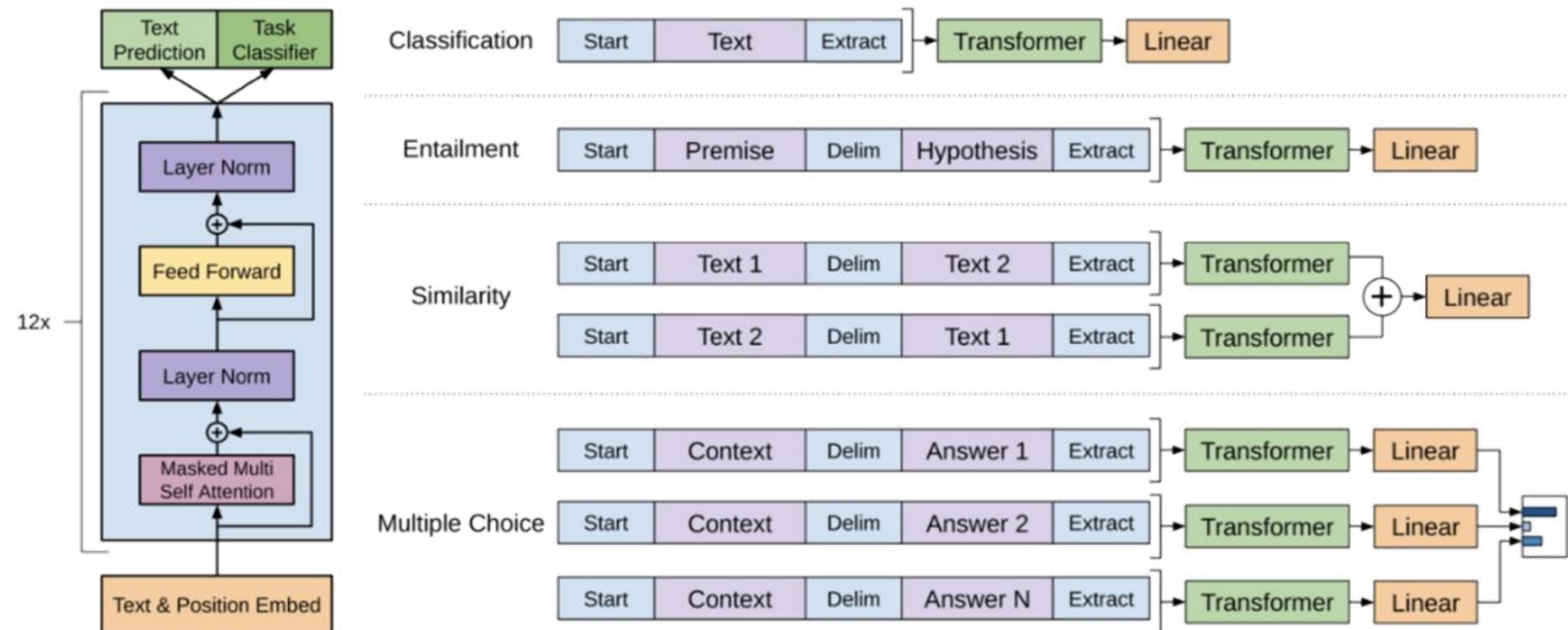
- Modern LLMs:
 - Transformer decoder blocks
 - Causal attention (no access to future tokens)
 - Left-to-right generation
- Contrast with BERT:
 - Bidirectional
 - Masked LM

GPT series

Generative Pretrained Transformer series of models from OpenAI

These are **decoder** transfer language models and can be used for generative tasks.

GPT (2018)



GPT-2 (Radford et al., 2019)

- Decoder-only
- Pretrained on more data:
 - WebText from Reddit (40GB)
- Good at Generation Tasks

Context (WebText test)	
Outfit 1: Typical	
This pairing was the first outfit I thought of when I bought the shoes. It's like a summer version of this Jake Grantham outfit; in fact, my shoes are close to the colors of his Nike Racers! Instead of a heavy Harris Tweed jacket and denim shirt, I'm wearing a cotton DB jacket and and a linen shirt. Both fabrics (in these colors) are an absolute must for summer, as they go with both dark and and light pants! As you can see, they pair wonderfully with the dark jeans and shoes. It's a pseudo menswear/prep outfit. Overall, this is a very casual outfit which is why I paired my sneakers with it. I'm not about wearing a full wool suit with sneakers (as GQ shows a lot) but I'm definitely open to keeping things casual, like this cotton DB. Casual fabrics are key to pulling off your sneakers in a dressed down menswear outfit. I'd even suggest to wear these sneakers with a khaki chino suit or a white linen suit. Just be sure to ditch the tie or wear a tee or polo; wearing a tie with sneakers is a bit too much	
Small model completion	GPT-2 completion
.	for me!
Outfit 2: High End	Outfit 2: Casual Jacket
This outfit was just in the eye of the storm. It's a hybrid of what ASICS does with streetwear (like J. Crew) and what J. Crew does with casual pieces that are a bit more traditional. Again, as I mentioned before, it's a low budget item and they can be very expensive. These will definitely be an issue if you're trying to work as a casual sneakermaker, and that's OK but don't hesitate if you want to try the lower price versions of their outfits! This version features a very low cut-cut neckline (which is not a bad thing), slightly longer legs, and a pair of long legs to keep things nice. My preferred color of color these items are blue-gray (which is my first colors, as they are a light white), or black (which is my favorite of my designs). As always, my pick of colors from this store have nothing to do with that other than the design. For a high end, I would recommend this to everyone!	I know that some people might be opposed to wearing sneakers in a turtleneck, but I wanted to be true to myself, so I went with a slim fitting turtleneck in a color more similar to my favorite color of the day. Overall, it's a classic turtleneck, with nothing too flashy. I did swap my shoes for these ASICS Tiger Killshots and I wish I hadn't did that. The shoe itself is very comfortable, however, I found that wearing them in these sneakers made my feet look a bit larger. I'd wear them with denim, chinos, whatever, but would never recommend wearing them alone. There are just too many variables involved.
Outfit 3: High Light	Outfit 3: Dress
This was a "high light" look while I had a chance to get some color. I think the color is based on	This is the final outfit I wore the ASICS Tiger Killshots. This version of the ASICS is a much more modern and streamlined version. The shoe has the most modern and minimal design that I can think of! With a lot of black, these sneakers look very cool. They'll definitely be very comfortable for summer. To me, wearing them with a suit is the easiest way to have a good pair of sneaker shoes. That's pretty much it! If

GPT-3 (Brown et al., 2020)

- Decoder-only
- Pretrained on even more data (45TB)

Dataset	Quantity (tokens)	Weight in training mix	Epochs elapsed when training for 300B tokens
Common Crawl (filtered)	410 billion	60%	0.44
WebText2	19 billion	22%	2.9
Books1	12 billion	8%	1.9
Books2	55 billion	8%	0.43
Wikipedia	3 billion	3%	3.4

- **Common Crawl**: web data over 8 years (metadata & text with filtering)
- **WebText2**: web pages from all outbound Reddit links from posts with 3+ upvotes
- **Books1 & Books2**: internet-based books corpora
- **Wikipedia**: English pages

GPT-3 "In-Context" Learning

– Zero-Shot

```
1 Translate English to French: ← task description
2 cheese => ..... ← prompt
```

– One-Shot

```
1 Translate English to French: ← task description
2 sea otter => loutre de mer ← example
3 cheese => ..... ← prompt
```

– Few-Shot

```
1 Translate English to French: ← task description
2 sea otter => loutre de mer ← examples
3 peppermint => menthe poivrée ← examples
4 plush girafe => girafe peluche ← examples
5 cheese => ..... ← prompt
```

Traditional Fine-Tuning



Autoregressive Models and Self-supervised Learning

- Autoregressive Models
 - Naturally generative
 - Can produce long coherent text
 - Supports in-context learning
- Self-supervised Learning
 - No manual labels needed.
 - Training signal comes from: (1) text itself; (2) next-token prediction
 - This scales easily to massive datasets.

Why Decoder-Only?

- Modern LLMs are predominantly decoder-only:
 - GPT-4o
 - Qwen3
 - Deepseek-R1
 - Llama4
 - ...
- Efficiency
 - Easier to scale and train
- Strong Pretraining
 - Excel in unsupervised pretraining
- Scalability
 - Handles longer contexts and complex tasks

What Is Pretraining?

- Pretraining:
 - Large-scale self-supervised training
 - Uses raw text (no human labels)
 - Learns general language modelling capability
- Goal: Learn to model the probability of text.

Language Modelling as Probability Estimation

- **Goal of pretraining:** Learn the probability distribution over text.
- Formally:

$$P(x_1, x_2, \dots, x_T)$$

$$P(x_1, \dots, x_T) = \prod_{t=1}^T P(x_t \mid x_{<t})$$

- **Key idea:** Model learns to predict each token given previous tokens.

Autoregressive Objective

- Autoregressive training objective:

$$\max \sum_t \log P(x_t \mid x_{<t})$$

- At each step:
 - See previous tokens (i.e. generation history)
 - Predict next token
 - Compute cross-entropy loss
 - Backpropagate gradients
- The model learns by minimising: Cross-entropy between predicted and true tokens.

What Does This Mean Practically?

- Input: “The capital of France is []”
- Target: “Paris”
- The model learns by:
 - Predicting next tokens
 - Over trillions of examples

Cross-Entropy Loss

- For each token:

$$\mathcal{L} = - \sum_i y_i \log p_i$$

- y_i = one-hot true token
 - p_i = predicted probability
- This penalises: Assigning low probability to correct token
- Intuition: Reward correct next-token predictions.

Training vs Inference

- Training:
 - Predict next token
 - Compare with ground truth
 - Update weights
- Inference:
 - Predict next token
 - Sample or choose max
 - Feed prediction back in

Limitation of the Objective

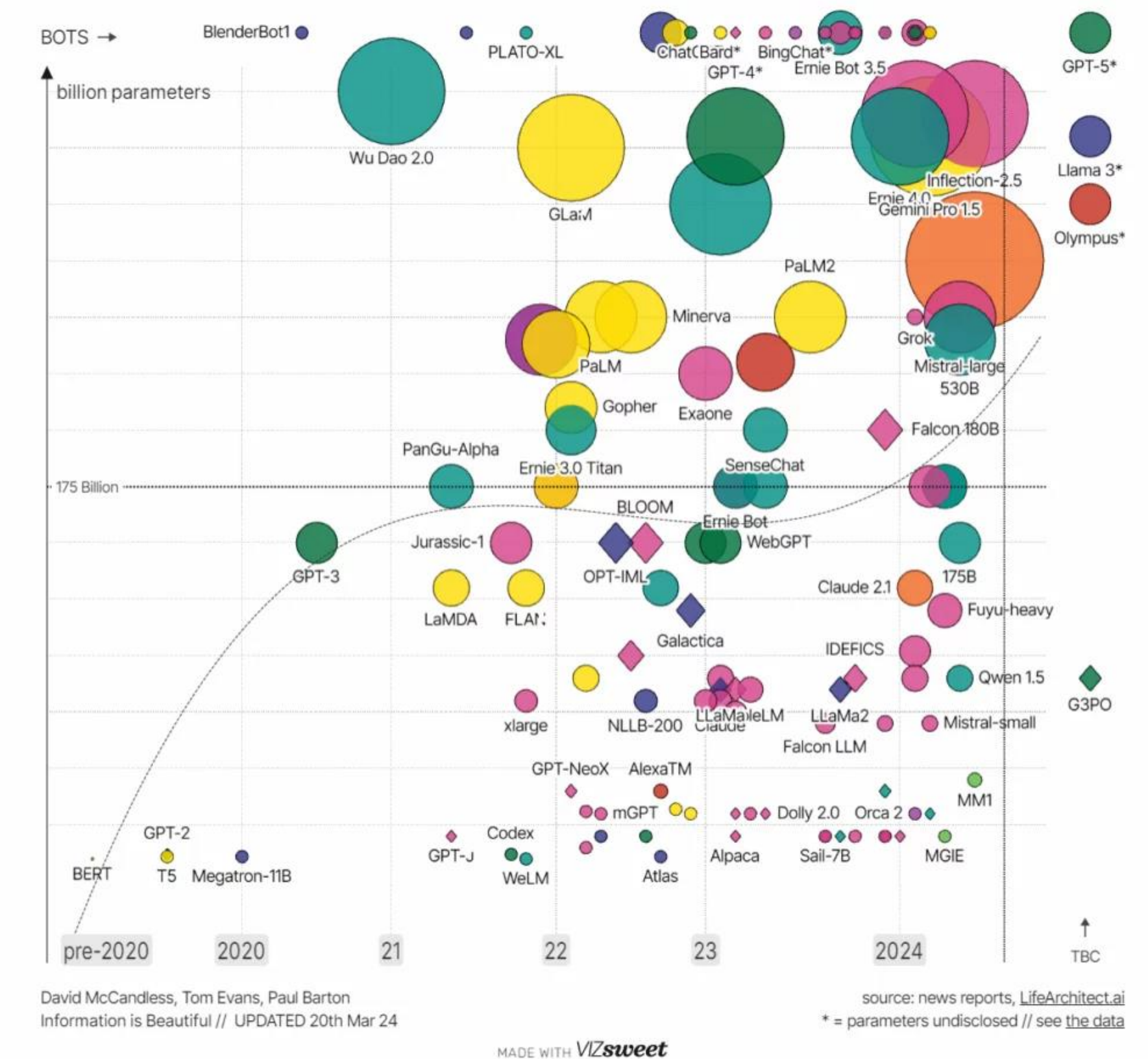
- Since objective is purely predictive, the model may:
 - Hallucinate plausible but false information
 - Repeat biases in data
 - Generate unsafe content
- Fundamentally, it optimises the likelihood of text, not correctness or other properties.

Training Data Scale

- Typical scale:
 - Billions to trillions of tokens
 - Web text, books, code, Wikipedia
- Data preprocessing is important
 - Deduplication
 - Filtering
 - Quality control

How large are “large” LMs?

- Today, we mostly talk about two camps of models:
 - Medium-sized models: BERT/RoBERTa models (100M or 300M), T5 models
 - (220M, 770M, 3B)
- “Very” large LMs: models of 100+ billion parameters
- Larger model sizes $\Rightarrow$ larger compute, more expensive during inference
- Different sizes of LMs have different ways to adapt and use them
 - Fine-tuning, zero-shot/few-shot prompting, in-context learning...
- Emergent properties arise from model scale
- Trade-off between model size and corpus size



The rise and rise of AI-based Large Language Models (LLMs) like GPT4, LaMDA, LLaMa, PaLM and Jurassic-2.

Compute Requirements

- Training can involve:
 - Thousands of GPUs / TPUs
 - Weeks to months of training
 - Distributed training
- Compute becomes the main constraint.
- Training vs inference cost
 - Training: expensive, done a few times (even once)
 - Inference: cheaper, but many times



Why LLMs

- The promise: one single model to solve many NLP tasks
- Emergent properties in LLMs

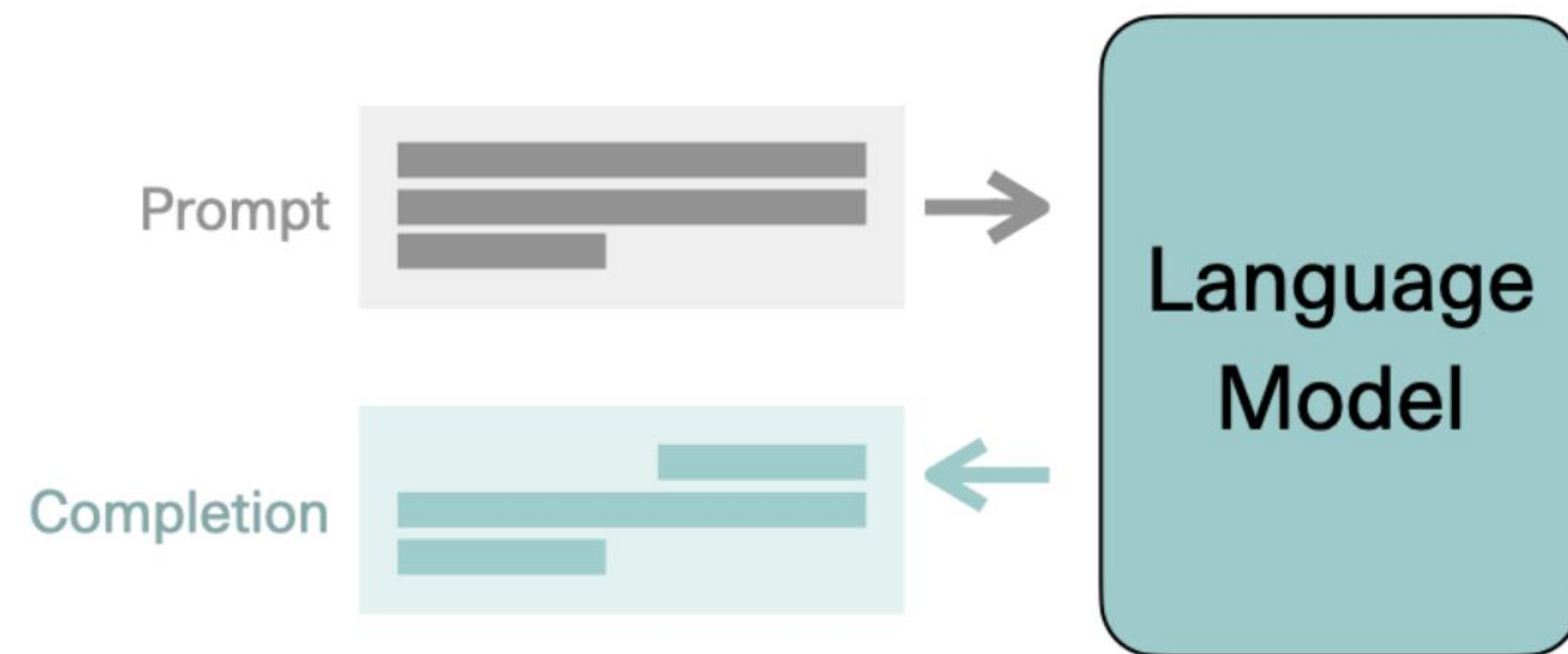
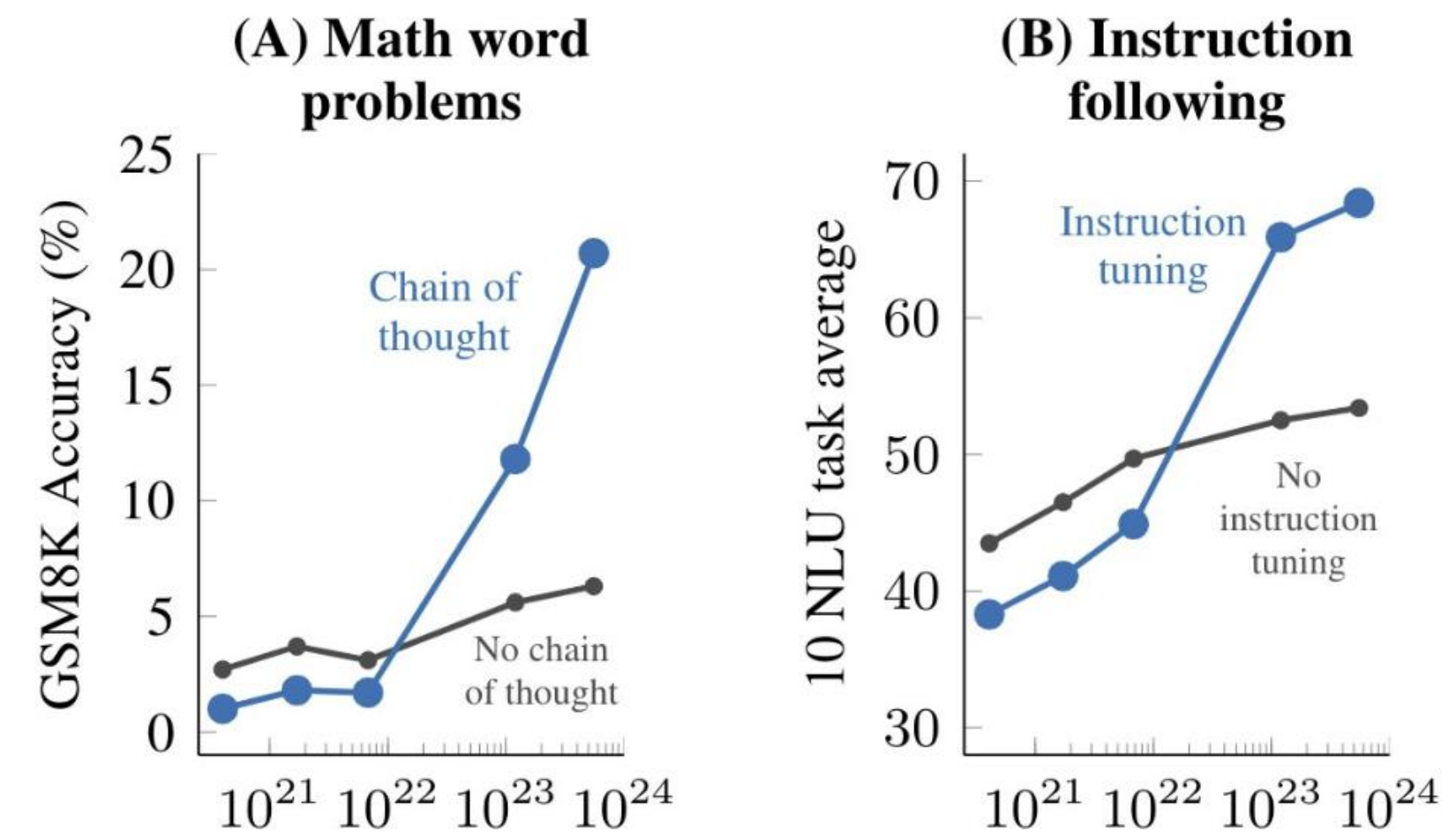


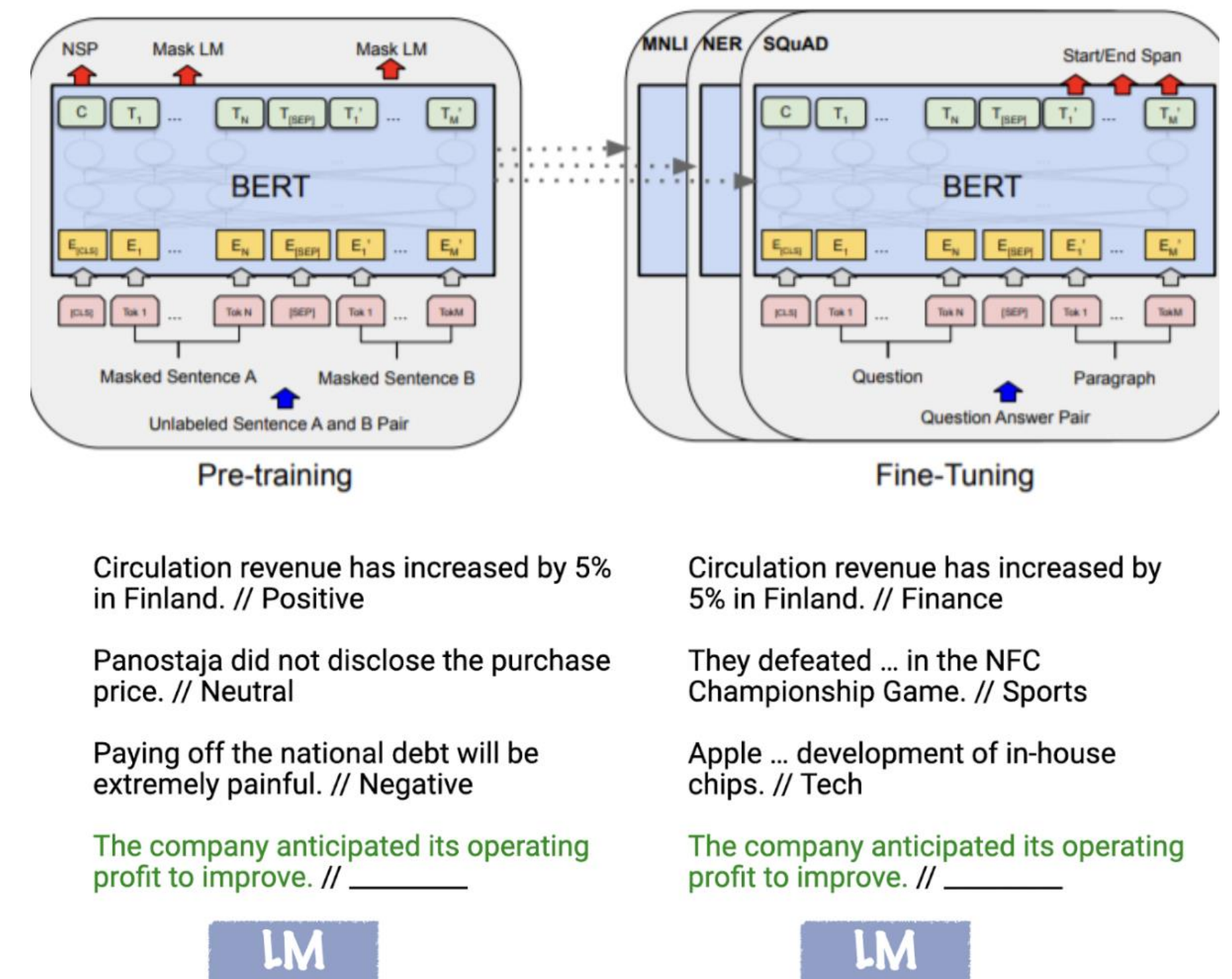
Image credit: Jay Alammar



(Wei et al., 2022)

Pre-training and adaption

- Pre-training: trained on huge
 - amounts of unlabeled text using “self-supervised” training objectives
- Adaptation: how to use a pre-trained model for your downstream task?
 - What types of NLP tasks (input and output formats)?
 - How many annotated examples do you have?
 - Approach: Fine-tuning, Prompting, Instruction tuning



<http://ai.stanford.edu/blog/understanding-incontext/>

The Dominant Architecture of Transformer-based LLMs

- GPT-3:
 - Decoder-only Transformer
 - Causal attention
 - LayerNorm, residual blocks, etc.
- Modern LLMs (LLaMA, Mistral, DeepSeek, etc.):
 - Same core architecture
 - Sometimes improved normalization or efficiency
 - Minor architectural refinements
 - But the core is the same.

Foundation vs Instruction Tuned Models

Foundation

- . Coherent, fluent generation
- . Completion and continuation
- . Vast world knowledge
- . Potentially harmful outputs
- . Difficult to control
- . Misaligned with our values

Instruction Tuned

- . Foundation ability and more
- . Controllable by prompt
- . Less need to fine-tune
- . Follows any command
- . Slight hit to performance
- . Sensitive to prompt template

Summary

- Pretraining
 - Large-scale self-supervised learning
 - Autoregressive next-token prediction
 - Trained on massive text corpora
- What It Produces
 - Fluent text
 - World knowledge correlations
 - In-context learning capability
 - Generative foundation model
- What It does NOT Guarantee
 - Instruction following
 - Helpfulness or safety
 - Truthfulness

